Guide: The Verified Residue-Coverage Tool

Collatz proof project

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What has been proved

Every positive Collatz orbit visits a number whose remainder on division by nine is two. Such visits occur after any specified time. This is a formalization of a known theorem of Monks and coauthors, not a new proof of the Collatz conjecture.

How the argument works

First move past any multiples of three. The remaining remainders are one, two, four, five, seven, and eight. If the orbit avoids two, the allowed remainder transitions force it eventually to stay at eight. An even number with remainder eight moves to remainder four, so this would require only odd steps forever. Our existing Lean parity argument rules that out for natural starting values.

How to use it in Lean

Import `Collatz.ResidueCoverage`. The theorem `exists_two_mod_nine` supplies a hitting time. The theorem `arbitrarily_late_two_mod_nine` also accepts a lower bound on the time. The theorem `reaches_one_of_two_mod_nine` reduces full convergence to convergence for positive seeds in that progression. Its convergence hypothesis still needs a proof.

What consecutive visits tell us

Between consecutive visits to the progression, at most six steps are even. The number of odd steps can still be arbitrarily large. If the accumulated coefficient is below one, however, the return takes at most sixteen steps. This bound is attained: 147440 next visits at 132860 after sixteen steps. Lean also proves that every such contracting return from a seed above two has an endpoint smaller than its starting value. The proof combines a global arithmetic bound with a kernel-checked finite table, so it applies to every seed, not just those tested numerically. Growing returns exist too: eleven next visits at twenty. We have not proved that every orbit eventually takes a contracting return. The new theorem identifies the possible contracting branches of this return map; it does not establish convergence of that map.

The missing connection

Earlier work proves that every hypothetical unbounded orbit contains starting points whose multiplicative coefficients never fall below one. It also restricts the smallest such starting point. The residue-coverage theorem does not say its chosen visit has that coefficient property. Two separately guaranteed kinds of visits need not occur at the same time. A proof using both results must establish that extra connection or use a different invariant.

Verification

From `lean/`, run `lake build Collatz.ResidueCoverage`. From the repository root, run `python3 analysis/audit_theorems.py --build` with the project's Lean executable on `PATH`. The selected audit checks the actual theorem statements and axiom dependencies; no additional axiom asserts the literature result.